Problem 1(b): QED Compton → QCD Gluon Compton

Full Derivation with Diagrams

PHYS 7502

1 QED Compton Diagrams and Labeling

Momentum Labeling Convention

Throughout this solution we use the labeling:

$$p_1 = e^-_{\text{in}}, \qquad p_2 = \gamma_{\text{in}}, \qquad p_3 = \gamma_{\text{out}}, \qquad p_4 = e^-_{\text{out}}$$

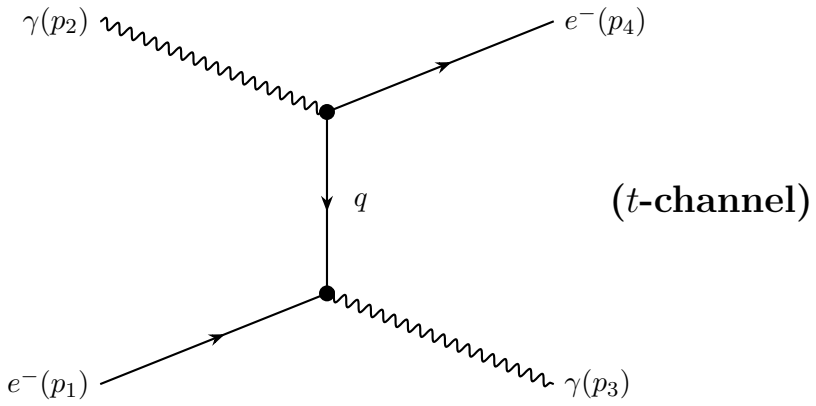
Mandelstam variables:

$$\hat{s} = (p_1 + p_2)^2, \qquad \hat{t} = (p_1 - p_3)^2, \qquad \hat{u} = (p_1 - p_4)^2$$

with $\hat{s} + \hat{t} + \hat{u} = 0$ in the massless limit.

There are two diagrams for QED Compton scattering, plus a third topology related by crossing to pair annihilation. Let's draw all three.

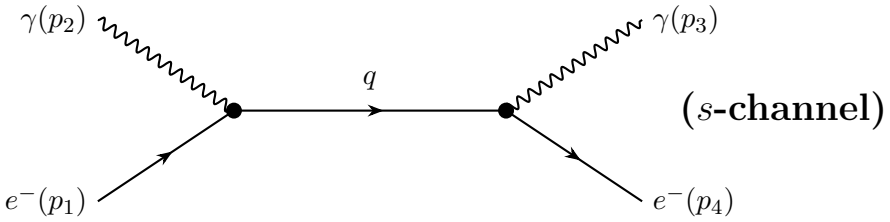
1.1 The t-Channel Diagram



$$q = p_1 - p_3 \implies q^2 = (p_1 - p_3)^2 = \hat{t}$$

The electron p_1 comes in, emits the outgoing photon p_3 at the bottom vertex, propagates as a virtual electron with momentum $q = p_1 - p_3$, then absorbs the incoming photon p_2 at the top vertex and exits as p_4.

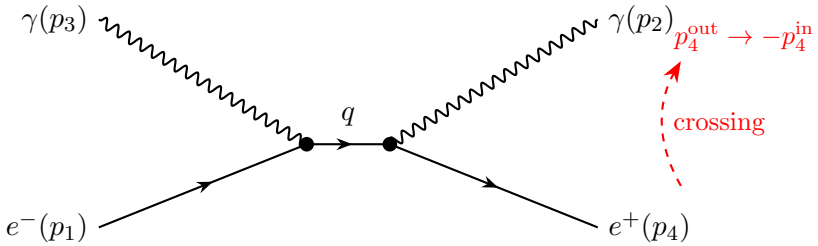
1.2 The s-Channel Diagram



$$q = p_1 + p_2 \implies q^2 = (p_1 + p_2)^2 = \hat{s}$$

The electron p_1 absorbs the incoming photon p_2 at the left vertex, propagates as a virtual electron with momentum $q = p_1 + p_2$, then emits the outgoing photon p_3 at the right vertex and exits as p_4.

1.3 The u-Channel: Pair Annihilation (Crossing Symmetry)



$$q^2 = (p_1 - p_4)^2 = \hat{u}$$

Pair annihilation topology (u-channel)

This diagram is obtained from the s-channel by **crossing**: swap the outgoing electron p_4 with an incoming positron $-p_4$. The virtual propagator carries momentum q with $q^2 = (p_1 - p_4)^2 = \hat{u}$. This topology corresponds to pair annihilation.

QED Compton: Channel Summary

Diagram	Propagator momentum	Denominator
t-channel	$q = p_1 - p_3$	$\hat{t} = (p_1 - p_3)^2$
s-channel	$q = p_1 + p_2$	$\hat{s} = (p_1 + p_2)^2$
u-channel (pair ann.)	$q = p_1 - p_4$	$\hat{u} = (p_1 - p_4)^2$

Only the s- and t-channel diagrams contribute to Compton scattering. The u-channel is a different physical process (pair annihilation), related by crossing symmetry.

Your QED result (spin/polarization-averaged, massless limit):

$$\boxed{\overline{|\mathcal{M}|^2}_{\text{QED}} = -2g_e^4\left(\frac{\hat{s}}{\hat{t}} + \frac{\hat{t}}{\hat{s}}\right) = \frac{-2g_e^4(\hat{s}^2 + \hat{t}^2)}{\hat{s}\hat{t}}} \tag{1}$$

$$\boxed{\left(\frac{d\hat{\sigma}}{d\hat{t}}\right)_{\text{QED}} = \frac{-g_e^4(\hat{s}^2 + \hat{t}^2)}{8\pi\hat{s}^3\hat{t}}} \tag{2}$$

2 QCD Gluon Compton: $qg \to qg$ Diagrams

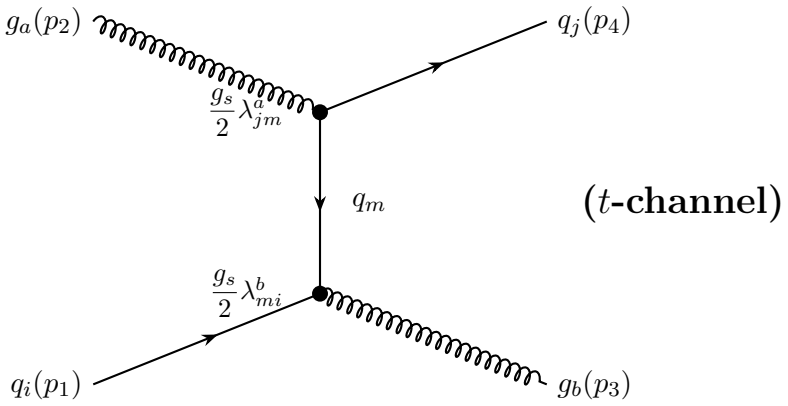
Same Labeling for QCD

$$p_1 = q_{\text{in}}, \qquad p_2 = g_{\text{in}}, \qquad p_3 = g_{\text{out}}, \qquad p_4 = q_{\text{out}}$$

Same Mandelstam variables: $\hat{s} = (p_1 + p_2)^2$, $\hat{t} = (p_1 - p_3)^2$, $\hat{u} = (p_1 - p_4)^2$.

With this labeling, the two quark-propagator diagrams are in the $\hat{s}$- and $\hat{t}$-channels — exactly matching the QED Compton topology. **No Mandelstam conversion needed.**

2.1 QCD t-Channel Diagram

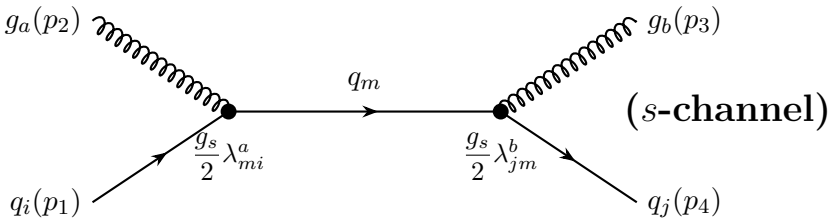


Propagator: $q^2 = (p_1 - p_3)^2 = \hat{t}$

Reading along the fermion line: quark i enters, emits gluon b at bottom vertex ($i \to m$: factor λ^b_{mi}), propagates, absorbs gluon a at top vertex ($m \to j$: factor λ^a_{jm}), exits as quark j.

$$\text{Color factor} = \frac{1}{4}\sum_m \lambda^a_{jm}\lambda^b_{mi} = \frac{1}{4}(\lambda^a\lambda^b)_{ji} \tag{3}$$

2.2 QCD s-Channel Diagram

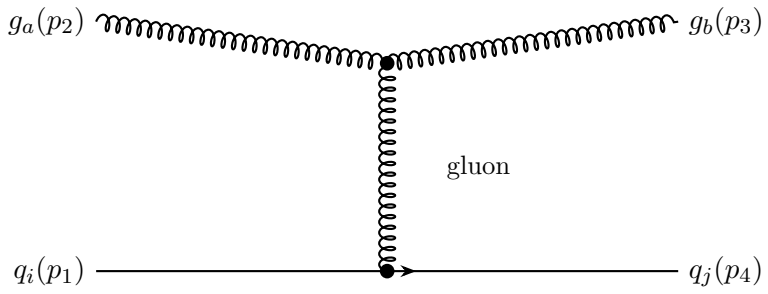


Propagator: $q^2 = (p_1 + p_2)^2 = \hat{s}$

Reading along the fermion line: quark i absorbs gluon a at left vertex ($i \to m$: factor λ^a_{mi}), propagates, emits gluon b at right vertex ($m \to j$: factor λ^b_{jm}), exits as quark j.

$$\text{Color factor} = \frac{1}{4}\sum_m \lambda^b_{jm}\lambda^a_{mi} = \frac{1}{4}(\lambda^b\lambda^a)_{ji} \tag{4}$$

2.3 The Excluded $\hat{u}$-Channel (Triple-Gluon Vertex)



Propagator: $q^2 = (p_1 - p_4)^2 = \hat{u}$

Triple-gluon vertex $\propto g_s f^{abc}$

EXCLUDED — no QED analog

Why the Channels Are $\hat{s}$ and $\hat{t}$ (Not $\hat{s}$ and $\hat{u}$)

With the labeling $p_1 = \text{fermion}_{\text{in}}$, $p_2 = \text{boson}_{\text{in}}$, $p_3 = \text{boson}_{\text{out}}$, $p_4 = \text{fermion}_{\text{out}}$:

- The **s-channel** has the fermion absorbing the incoming boson first: $q = p_1 + p_2$, so $q^2 = \hat{s}$.
- The **t-channel** has the fermion emitting the outgoing boson first: $q = p_1 - p_3$, so $q^2 = \hat{t}$.
- The **u-channel** would require a three-boson vertex ($q = p_1 - p_4$). This doesn't exist for photons (QED) and is excluded for gluons per the problem instructions.

In many textbooks, the outgoing particles are labeled differently (e.g., $p_3 = e^-_{\text{out}}$, $p_4 = \gamma_{\text{out}}$), which swaps $\hat{t} \leftrightarrow \hat{u}$ and gives propagator channels $\hat{s}$ and $\hat{u}$. The physics is identical — only the labeling differs.

3 The QCD Conversion: All Steps

3.1 Step 1: Write the Factorized QCD Amplitudes

The QCD vertex factor is $-\frac{ig_s}{2}\lambda^a_{ji}\gamma^\mu$ (Griffiths Eq. 9.17). The λ^a matrix acts on color space; γ^μ acts on spinor space. They **factorize**:

$$\mathcal{M}_t = \frac{g_s^2}{4}\,(\lambda^a\lambda^b)_{ji} \;\times\; D_t \qquad (t\text{-channel}) \tag{5}$$

$$\mathcal{M}_s = \frac{g_s^2}{4}\,(\lambda^b\lambda^a)_{ji} \;\times\; D_s \qquad (s\text{-channel}) \tag{6}$$

where D_t and D_s are the Dirac structures — identical to QED Compton with $e^2 \to 1$:

$$D_t = \frac{-1}{\hat{t}}\,\bar{u}(p_4)\not{\epsilon}_2(\not{p}_1 - \not{p}_3)\not{\epsilon}_3^*\,u(p_1), \qquad D_s = \frac{-1}{\hat{s}}\,\bar{u}(p_4)\not{\epsilon}_3^*(\not{p}_1 + \not{p}_2)\not{\epsilon}_2\,u(p_1) \tag{7}$$

Why the Color Matrices Are Different for Each Diagram

In the t-channel, the quark emits gluon b first (factor λ^b_{mi}), then absorbs gluon a (factor λ^a_{jm}), giving $(\lambda^a\lambda^b)_{ji}$.
In the s-channel, the quark absorbs gluon a first (factor λ^a_{mi}), then emits gluon b (factor λ^b_{jm}), giving $(\lambda^b\lambda^a)_{ji}$.
These differ because $\lambda^a\lambda^b \neq \lambda^b\lambda^a$ — the Gell-Mann matrices do not commute (SU(3) is non-Abelian).

3.2 Step 2: Square and Sum Over Spins

$$|\mathcal{M}|^2 = \frac{g_s^4}{16}\left[|(\lambda^a\lambda^b)_{ji}|^2|D_t|^2 + |(\lambda^b\lambda^a)_{ji}|^2|D_s|^2 + 2\,\mathrm{Re}\Big((\lambda^a\lambda^b)^*_{ji}(\lambda^b\lambda^a)_{ji}\,D_t^*D_s\Big)\right] \tag{8}$$

Spin/polarization averaging (factor $\frac{1}{4}$: 2 quark spins $\times$ 2 gluon polarizations):

$$\frac{1}{4}\sum_{\text{spins,pols}}|D_t|^2 = -\frac{2\hat{s}}{\hat{t}} \qquad (\text{from your QED } t\text{-channel calculation}) \tag{9}$$

$$\frac{1}{4}\sum_{\text{spins,pols}}|D_s|^2 = -\frac{2\hat{t}}{\hat{s}} \qquad (\text{from your QED } s\text{-channel calculation}) \tag{10}$$

$$\frac{1}{4}\sum_{\text{spins,pols}} D_t^*D_s = 0 \qquad (\text{no spin interference at } m=0) \tag{11}$$

Why the Spin Interference Vanishes

In the massless limit, helicity is conserved at each QED/QCD vertex. The s-channel and t-channel diagrams correspond to different helicity configurations of the fermion–boson system. Different helicity amplitudes are orthogonal, so their interference vanishes.
Consequence: Even though the color factors $(\lambda^a\lambda^b)_{ji}$ and $(\lambda^b\lambda^a)_{ji}$ are different, the color interference term multiplies zero and drops out. We only need $|t\text{-channel}|^2$ and $|s\text{-channel}|^2$ color factors.

3.3 Step 3: Color Factor — Full Derivation

We average over initial colors and sum over final colors:

$$\text{Average:} \quad \frac{1}{N_c} \cdot \frac{1}{N_c^2 - 1} = \frac{1}{3} \cdot \frac{1}{8} = \frac{1}{24} \tag{12}$$

$$\text{Sum:} \quad \sum_{i,j=1}^{3} \sum_{a,b=1}^{8} \tag{13}$$

The color factor for $|t\text{-channel}|^2$ (the s-channel is identical by $a \leftrightarrow b$ symmetry):

$$\mathcal{C} = \frac{1}{24} \sum_{i,j,a,b} \left| (\lambda^a \lambda^b)_{ji} \right|^2 \tag{14}$$

Step 3a. Use Hermiticity ($\lambda^{a\dagger} = \lambda^a$):

$$\left| (\lambda^a \lambda^b)_{ji} \right|^2 = (\lambda^a \lambda^b)_{ji} \, (\lambda^a \lambda^b)^*_{ji} = (\lambda^a \lambda^b)_{ji} \, (\lambda^b \lambda^a)_{ij} \tag{15}$$

Step 3b. Sum over i, j produces a trace:

$$\sum_{i,j} (\lambda^a \lambda^b)_{ji} (\lambda^b \lambda^a)_{ij} = \mathrm{Tr}\Big[\lambda^a \lambda^b \lambda^b \lambda^a\Big] \tag{16}$$

So:

$$\mathcal{C} = \frac{1}{24} \sum_{a,b} \mathrm{Tr}\Big[\lambda^a \lambda^b \lambda^b \lambda^a\Big] \tag{17}$$

Step 3c. Evaluate inner sum using the quadratic Casimir:

$$\sum_{b=1}^{8} \lambda^b \lambda^b = 4 C_F \, \mathbb{I}_3 = \frac{16}{3} \, \mathbb{I}_3 \qquad \text{where} \quad C_F = \frac{N_c^2 - 1}{2 N_c} = \frac{4}{3} \tag{18}$$

Explicit Check of the Casimir

$\sum_b (\lambda^b \lambda^b)_{11} = \sum_b \sum_k |\lambda^b_{1k}|^2$:

λ^b	nonzero $\lvert\lambda^b_{1k}\rvert^2$	value
λ^1	$\lvert\lambda^1_{12}\rvert^2 = \lvert 1\rvert^2$	1
λ^2	$\lvert\lambda^2_{12}\rvert^2 = \lvert -i\rvert^2$	1
λ^3	$\lvert\lambda^3_{11}\rvert^2 = \lvert 1\rvert^2$	1
λ^4	$\lvert\lambda^4_{13}\rvert^2 = \lvert 1\rvert^2$	1
λ^5	$\lvert\lambda^5_{13}\rvert^2 = \lvert -i\rvert^2$	1
λ^6	—	0
λ^7	—	0
λ^8	$\lvert\lambda^8_{11}\rvert^2 = \lvert 1/\sqrt{3}\rvert^2$	1/3
Total		16/3 ✓

Step 3d. Substitute:

$$\mathcal{C} = \frac{1}{24}\sum_a \mathrm{Tr}\left[\lambda^a \cdot \frac{16}{3}\mathbb{I} \cdot \lambda^a\right] = \frac{1}{24}\cdot\frac{16}{3}\sum_a \mathrm{Tr}(\lambda^a\lambda^a) \tag{19}$$

Step 3e. Use $\mathrm{Tr}(\lambda^a\lambda^b) = 2\delta^{ab}$:

$$\sum_a \mathrm{Tr}(\lambda^a\lambda^a) = \sum_a 2 = 2\times 8 = 16 \tag{20}$$

Step 3f. Final color factor:

$$\mathcal{C} = \frac{1}{24}\cdot\frac{16}{3}\cdot 16 = \frac{256}{72} = \frac{32}{9} \tag{21}$$

Color Factor Result

$$\boxed{\mathcal{C}_t = \mathcal{C}_s = \frac{32}{9}} \tag{22}$$

Anatomy: $\dfrac{32}{9} = \underbrace{\dfrac{1}{3}}_{\text{avg } q_{\text{in}}} \times \underbrace{\dfrac{1}{8}}_{\text{avg } g_{\text{in}}} \times \underbrace{\dfrac{256}{3}}_{\sum_{i,j,a,b} |(\lambda^a\lambda^b)_{ji}|^2}$

3.4 Step 4: Assemble $\overline{|\mathcal{M}|^2}_{\mathbf{QCD}}$

Combining coupling ($g_s^4/16$), spin ($-2\hat{s}/\hat{t}$ and $-2\hat{t}/\hat{s}$), and color ($32/9$):

$$\begin{aligned}\overline{|\mathcal{M}|^2}_{\mathrm{QCD}} &= \frac{g_s^4}{16}\times\frac{32}{9}\times\left[-2\left(\frac{\hat{s}}{\hat{t}}+\frac{\hat{t}}{\hat{s}}\right)\right]\\ &= \frac{2g_s^4}{9}\times\left[-2\left(\frac{\hat{s}}{\hat{t}}+\frac{\hat{t}}{\hat{s}}\right)\right]\\ &= \frac{-4g_s^4(\hat{s}^2+\hat{t}^2)}{9\,\hat{s}\hat{t}}\end{aligned} \tag{23}$$

Cross-check: Compare with QED (1):

$$\frac{\overline{|\mathcal{M}|^2}_{\mathrm{QCD}}}{\overline{|\mathcal{M}|^2}_{\mathrm{QED}}} = \frac{-4g_s^4/(9\hat{s}\hat{t})}{-2g_e^4/(\hat{s}\hat{t})} = \frac{2g_s^4}{9g_e^4} = \frac{2\alpha_s^2}{9\alpha^2} \qquad \checkmark \tag{24}$$

3.5 Step 5: Cross Section

Using $g_s^2 = 4\pi\alpha_s \implies g_s^4 = 16\pi^2\alpha_s^2$:

$$\overline{|\mathcal{M}|^2}_{\mathrm{QCD}} = \frac{-64\pi^2\alpha_s^2(\hat{s}^2+\hat{t}^2)}{9\,\hat{s}\hat{t}} \tag{25}$$

The invariant cross section $d\hat{\sigma}/d\hat{t} = \overline{|\mathcal{M}|^2}/(16\pi\hat{s}^2)$:

$$\frac{d\hat{\sigma}}{d\hat{t}} = \frac{-64\pi^2\alpha_s^2(\hat{s}^2+\hat{t}^2)}{9\times 16\pi\times \hat{s}^3\hat{t}} \tag{26}$$

Final Answer: $qg \to qg$ (Two Diagrams)

$$\boxed{\left.\frac{d\hat{\sigma}}{d\hat{t}}\right|_{2\text{ diag}} = \frac{-4\pi\alpha_s^2(\hat{s}^2+\hat{t}^2)}{9\,\hat{s}^3\hat{t}}} \tag{27}$$

Equivalently:

$$\boxed{\left.\frac{d\hat{\sigma}}{d\hat{t}}\right|_{\text{QCD}} = \frac{2\alpha_s^2}{9\alpha^2}\times\left.\frac{d\hat{\sigma}}{d\hat{t}}\right|_{\text{QED}}} \tag{28}$$

4 Summary

Complete Pipeline

Your QED Compton result (propagator channels $\hat{s}$ and $\hat{t}$):

$$\left.\frac{d\hat{\sigma}}{d\hat{t}}\right|_{\text{QED}} = \frac{-g_e^4(\hat{s}^2+\hat{t}^2)}{8\pi\hat{s}^3\hat{t}}$$

Coupling replacement: $g_e^4 \to g_s^4/16$ (two QCD vertices, each $g_s/2$)
Color factor: $\mathcal{C} = 32/9$ (derived via $\text{Tr}[\lambda^a\lambda^b\lambda^b\lambda^a]$)
Combined multiplier: $\dfrac{g_s^4/16}{g_e^4} \times \dfrac{32}{9} = \dfrac{2\alpha_s^2}{9\alpha^2}$
Final QCD result:

$$\boxed{\left.\frac{d\hat{\sigma}}{d\hat{t}}\right|_{\text{QCD}} = \frac{-4\pi\alpha_s^2(\hat{s}^2+\hat{t}^2)}{9\,\hat{s}^3\hat{t}}}$$

Mandelstam conversion needed? NO. Same labeling (p_1=fermion$_{\text{in}}$, p_2=boson$_{\text{in}}$, p_3=boson$_{\text{out}}$, p_4=fermion$_{\text{out}}$) gives the same propagator channels ($\hat{s}$ and $\hat{t}$) for both QED and QCD.

What the Third Diagram ($\hat{u}$-channel) Adds

The full gauge-invariant $qg \to qg$ result includes the triple-gluon vertex diagram with propagator $1/\hat{u}$. This diagram has:

- A different Dirac structure (gluon propagator, not quark)
- Color factor involving f^{abc} (structure constants)
- No QED analog (photons don't self-interact)

The complete result (Halzen & Martin) contains an additional $\hat{u}$-channel contribution. This will be discussed in class.